

Inference-Time Scaling in Latent Space: Recurrent Non-Verbal Reasoning

Track Submission: DataForge x Pathway — *Explain the Frontier*
Artifact: <https://github.com/TheShriHari/Dataforge-Submission>
Topic: Test-Time Compute Scaling & Recurrent Latent Reasoning (BDH / BDH-CQ)

1. Central Claim

"Increasing the number of recurrent latent computation steps at inference time can improve reasoning accuracy without producing any additional natural-language reasoning tokens — but gains diminish and each extra step costs more compute/time."

2. Design Motivation: Latent Iteration vs. Verbalized Chain-of-Thought

Current test-time scaling in Large Language Models primarily relies on autoregressive Chain-of-Thought (CoT) prompting or beam-search sampling (Snell et al., 2024, [arXiv:2408.03314](https://arxiv.org/abs/2408.03314)). While effective, verbalized CoT suffers from severe architectural constraints:

- Context & Memory Bloat:** Every verbalized reasoning step appends new tokens to the context window, causing $O(N^2)$ self-attention cost and quadratic Key-Value (KV) cache memory expansion.
- Syntactic Fragility & Semantic Drift:** Verbal reasoning requires tokens to adhere to natural-language grammar; an early syntactic or hallucinations flaw permanently corrupts all downstream generated tokens.
- Difference from Standard Forward Passes:** A standard feedforward pass (e.g., standard Transformer layer stack or feedforward CNN) applies a static sequence of layers where total compute is strictly bound to parameter depth L . In contrast, **recurrent latent iteration** feeds the hidden state back into a shared-weight core for K discrete iterations: $h_k = \text{Core}(h_{k-1}, x)$ for $k \in [1, K]$. This decouples inference computation budget from static parameter size, preserving a constant tensor memory footprint ($h \in \mathbb{R}^{C \times H \times W}$) with zero emitted text tokens.

3. Technical Mechanism & Measured Trade-Offs

Our implementation investigates this mechanism on 2D maze navigation, where long-range spatial reachability requires non-local computational depth. Using a 123,713-parameter shared Convolutional GRU (ConvGRU) core (`ml/model.py`), each recurrence step k expands the effective spatial receptive field by 2 cells.

Measured across $N=100$ held-out 15×15 synthetic mazes (Seed 999, `results/accuracy_by_effort.json`):

- Steep Early Accuracy Phase ($K = 1 \rightarrow 10$):**
 - Exact Solve Rate jumps from **1.0%** ($K = 1$, 0.666 ms latency) to **44.0%** ($K = 5$, 2.911 ms), reaching **81.0%** at $K = 10$ (6.155 ms).
 - Continuous Path IoU rises from **55.94%** to **96.61%**, demonstrating rapid convergence as information propagates through corridors.
- Empirical Diminishing Returns ($K = 10 \rightarrow 20$):**
 - Pushing K from 10 to 20 yields only a **+7.0%** increase in solve rate (81.0% $\rightarrow$ 88.0%) and plateaued Path IoU (96.61% $\rightarrow$ 98.13%).
 - Meanwhile, inference latency more than doubles from **6.155 ms to 12.884 ms/maze**. Every marginal gain after the topological path is discovered incurs strictly linear compute cost for near-zero returns.
- Training Horizon & Generalization:** $K = 16\text{--}20$ extends beyond the $K \sim \text{Uniform}(1, 15)$ training curriculum (`ml/train.py`). The model generalizes to these values because the recurrent core is weight-shared and has no explicit step-count parameter, but this was not part of the training distribution.

4. Multi-Dimensional Paradigm Comparison

Dimension	Toy Latent Maze Solver (Our Artifact)	BDH-CQ Foundation Model (Kosowski et al., 2026)	Autoregressive Search / CoT (Snell et al., 2024)
Domain & Scale	2D Spatial Pathfinding (15×15), 123K params.	ARC-AGI-1 Abstract Reasoning, $\sim 150M$ params.	Symbolic Math & Language Benchmarks, $> 7B$ params.
Effort vs. Accuracy	Solves scale from 1.0% ($K = 1$) to 88.0% ($K = 20$) ; steep early gains, plateauing at $K \geq 11$.	Pass@2 scales from 21.0% (Low) to 29.5% (High) on ARC-AGI-1 (Table 5).	Accuracy scales with sample budget N and search beam depth.
Inference Hardware Cost	Latency scales linearly from 0.666 ms ($K = 1$) to 12.884 ms ($K = 20$).	Low effort cuts hardware compute cost by 22.0% (Medium cuts by 11.0%) vs. High effort.	Exponentially growing KV-cache and sequence compute cost ($O(N^2)$).
Memory Footprint	Constant $O(1)$ tensor state ($\mathbb{R}^{48 \times 15 \times 15}$).	Constant latent state manifold during recurrent updates.	Linear-to-quadratic memory expansion per generated token.
Observability	Continuous probability readout via 1×1 projection head.	Continuous hidden activations (non-symbolic latent space).	High: Human-readable natural language reasoning transcripts.

5. Architectural Roles: BDH vs. BDH-CQ

- To avoid conflation, the two literature systems are distinguished:
- The Dragon Hatchling (BDH)** (Kosowski et al., 2025, [arXiv:2509.26507](https://arxiv.org/abs/2509.26507)): A biologically-inspired, post-transformer foundational architecture that establishes recurrent latent computation and non-autoregressive parallel processing across continuous representations.
 - BDH-CQ** (Kosowski et al., 2026, [arXiv:2608.09888](https://arxiv.org/abs/2608.09888)): A specialized system built within the BDH architectural family incorporating in-context demonstration learning ("Context Queries") and explicit test-time effort modulation across LOW, MEDIUM, and HIGH budgets without emitting scratchpad text tokens.

6. Evidence Classification: Strengths vs. Weaknesses

- Our Experimental Results:** Classified as **own toy implementation, empirically measured, not independently audited**. Evaluated on procedural synthetic mazes; while numerically reproducible and rigorously benchmarked with deterministic seeds, it operates on a restricted 2D spatial domain rather than general multi-modal reasoning.
- BDH-CQ External Baseline:** The ARC-AGI-1 benchmark score (29.5% pass@2) is **developer-reported, partially independently audited by Bielik/NYU for the headline score only**. The internal scaling ablation (Table 5: 21.0% / 27.0% / 29.5% across effort tiers) remains developer-reported by Pathway's research group and has not been independently reproduced at scale.

7. Primary Limitation & Open Question

While recurrent latent scaling eliminates token-generation overhead and memory bloat, it introduces a fundamental interpretability bottleneck: **latent reasoning lacks explicit symbolic intermediate auditability**. If a model fails at step $K = 4$, diagnosing whether the failure stemmed from latent channel saturation, gradient vanishing across recurrent unrolling, or feature drift requires probing classifiers rather than reading an English scratchpad. Developing standardized, zero-overhead probing techniques for continuous recurrent latent trajectories remains a crucial open research question.